



Correlation between Sports Event Tiers, Medal Counts, and Public Attention across Various Sports

Wang Chao-An
Li Chen-Hsun
Lin Yu-Chen
Hsu Chuan-Chi
Chen Yi-Hong

Research Motivation

Our research aims to explore whether there is a correlation between the competitive performance of Taiwanese athletes in major tournaments and the level of domestic discussion regarding those sports. By analyzing the relationship between the two, we hope to understand whether athletic achievements—aside from bringing glory to the nation—can influence domestic attention toward the sport and positively impact the domestic atmosphere regarding sports and health.

Data Sources

We selected highly active "Sports Specialty Boards" from PTT (Taiwan's largest bulletin board system) with sufficient data volume: Table Tennis, Badminton, and Swimming.

	Title	Date	Time	URL	Content
徵求	高雄徵游泳教練	2023-08-25	23-04	https://www.ptt.cc/bbs/Sport/001	我覺得可以先調整自由式動作1000
徵求	9/24日月潭名額1位	2023-08-31	18-38	https://www.ptt.cc/bbs/Sport/001	推 JosephChen: 已私114.137.242.216 09/01 10:14
新聞	1500名泳將齊聚台中爭冠 全國總決賽	2023-09-01	15-54	https://www.ptt.cc/bbs/Sport/001	推 marathons: 多板被水桶,到處流澀洗文貼簽名權
新聞	時薪上看350元還是乏人問津 桃津	2023-09-02	21-53	https://www.ptt.cc/bbs/Sport/001	推 marathons: 多板被水桶,到處流澀洗文貼簽名權
新聞	游泳／王冠閏、王星鳴有望終結	2023-09-04	09-19	https://www.ptt.cc/bbs/Sport/001	推 marathons: 多板被水桶,到處流澀洗文貼簽名權
徵求	高雄一對一游泳教練	2023-09-07	21-18	https://www.ptt.cc/bbs/Sport/001	推 ivydowney: 已站內~ 101.9.190.42 09/09 21:5

Data Format

We compiled the dates and medal records of Taiwanese athletes for major events such as the Olympics, Asian Games, and World Championships. Sources included news reports and the websites of various sports associations.

```
# general
olymp_dates = ["2008-08", "2012-07", "2016-08", "2021-07"]
asia_dates = ["2008-08", "2010-12", "2014-09", "2018-09", "2023-09"]
```

Preprocess

- Classified posts based on their titles.
- Calculated volume on a monthly basis.
- Applied the HP Filter to eliminate the long-term trend of declining user activity.

Methodology

Model 1 (Event Effect on Article Counts):

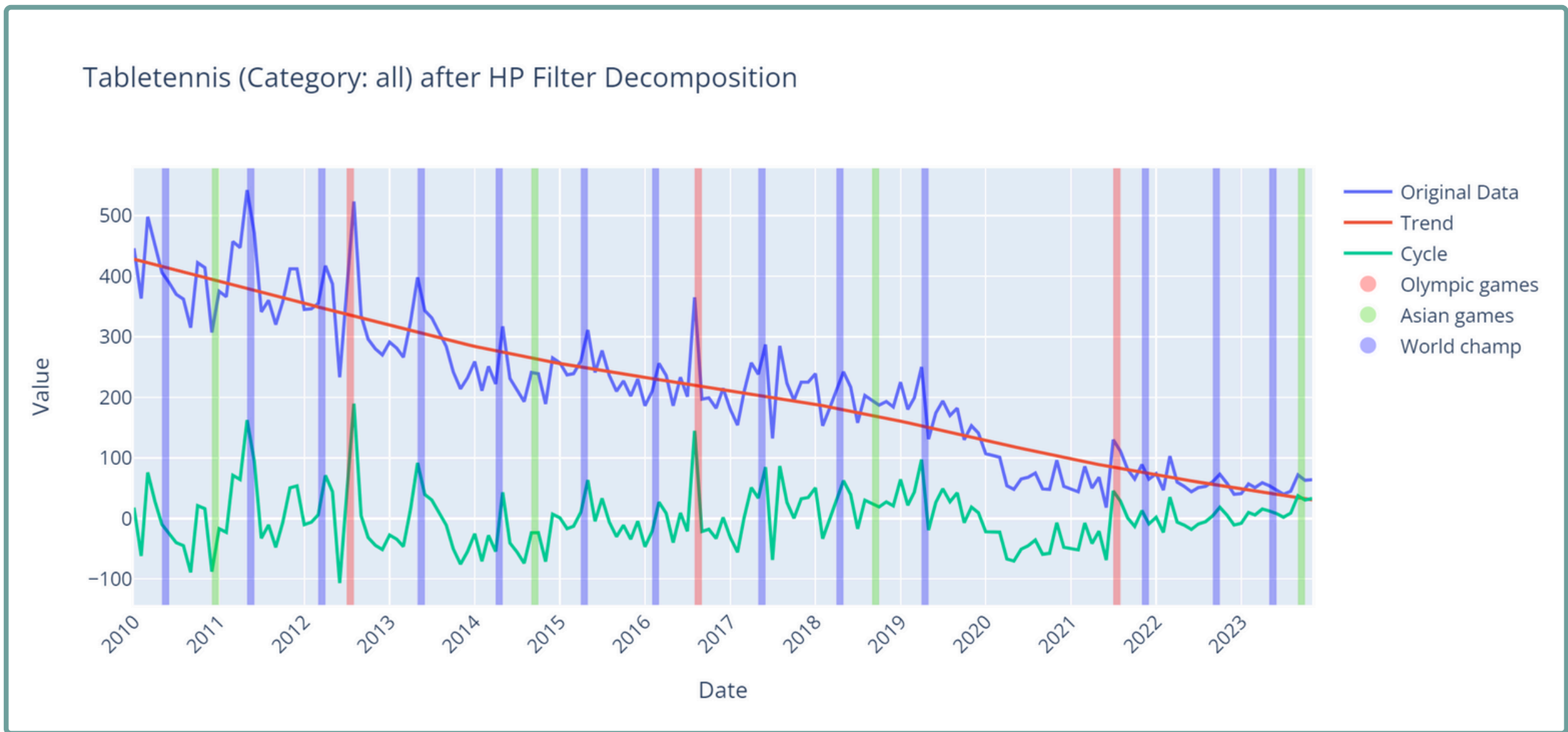
$$\text{Articles}_{\text{sport},t} = \beta_1 + \beta_2 \text{Olympics}_{\text{sport},t} + \beta_3 \text{Asiangames}_{\text{sport},t} + \beta_4 \text{WorldChampionships}_{\text{sport},t}$$

Model 2 (Medal Effect on Article Counts):

$$\text{Articles}_{\text{sport},t} = \beta_1 + \beta_2 \text{Olympics_awards}_{\text{sport},t} + \beta_3 \text{Asiangames_awards}_{\text{sport},t} + \beta_4 \text{WorldChampionships_awards}_{\text{sport},t}$$

Discussion & Results

Table Tennis



Regression of Model 1 (Monthly Basis, Event)

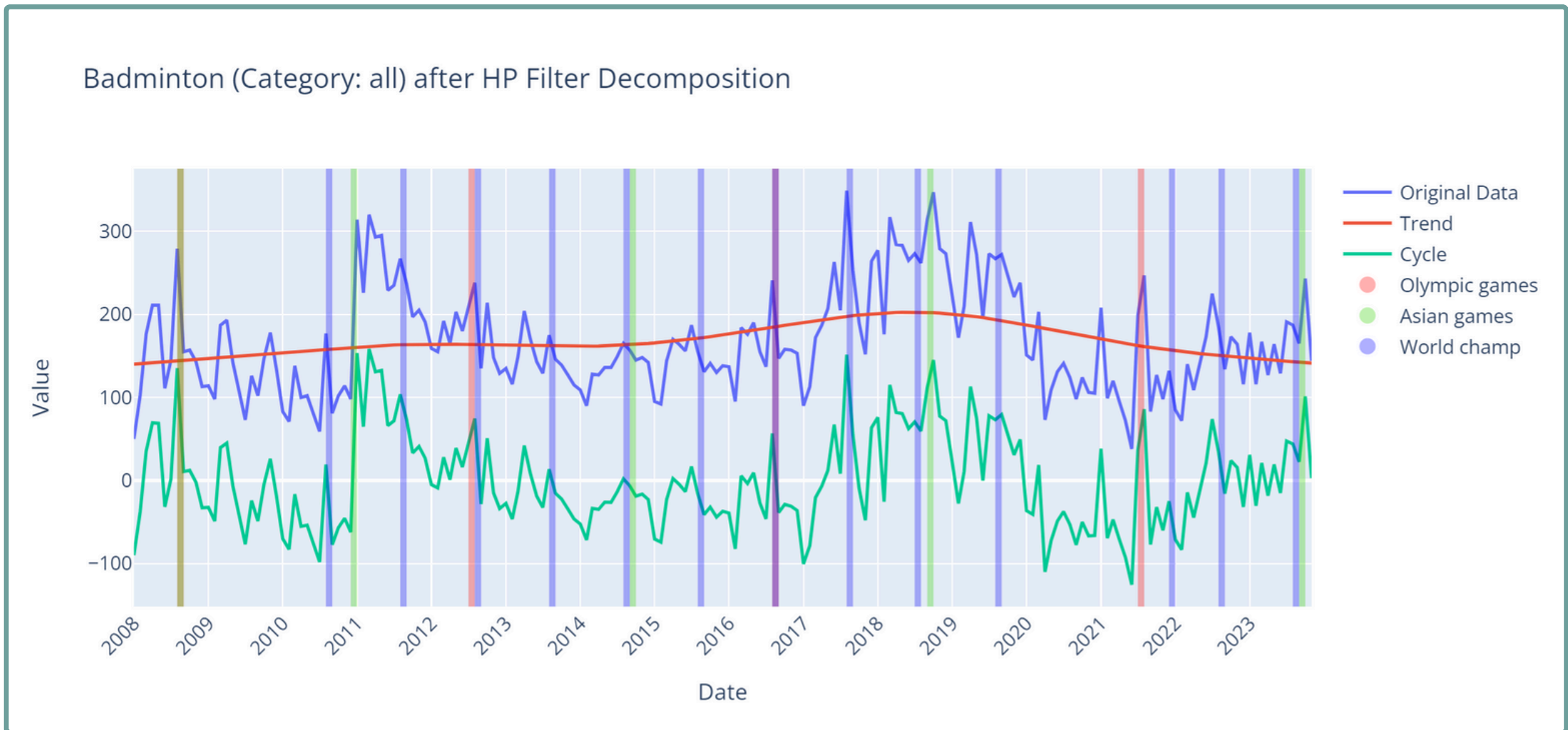
	coef	std err	t	P> t	[0.025	0.975]
Intercept	-5.8360	3.073	-1.899	0.059	-11.904	0.232
Olympics_tabletennis_dummy	98.3522	16.887	5.824	0.000	65.007	131.697
AsiaGame_tabletennis_dummy	14.1703	18.817	0.753	0.453	-22.987	51.328
WC_tabletennis_dummy	35.5141	11.150	3.185	0.002	13.497	57.531

Regression of Model 2 (Monthly Basis, Medal)

	coef	std err	t	P> t	[0.025	0.975]
Intercept	-1.5887	3.258	-0.488	0.626	-8.022	4.845
Olympics_table_prize	56.8336	29.160	1.949	0.053	-0.746	114.413
AsiaGame_table_prize	8.3165	18.431	0.451	0.652	-28.077	44.710
WC_table_prize	25.3382	18.614	1.361	0.175	-11.418	62.094

Both the Olympics and World Championships have a positive impact on table tennis discussion levels, showing significance at the 95% confidence level. Additionally, the medal count of our national team at the Olympics is significant at the 90% confidence level. In this sport, the public pays more attention to high-tier tournaments and focuses on the competition as a whole, rather than solely on whether our athletes win medals.

Badminton



Regression of Model 1 (Monthly Basis, Event)

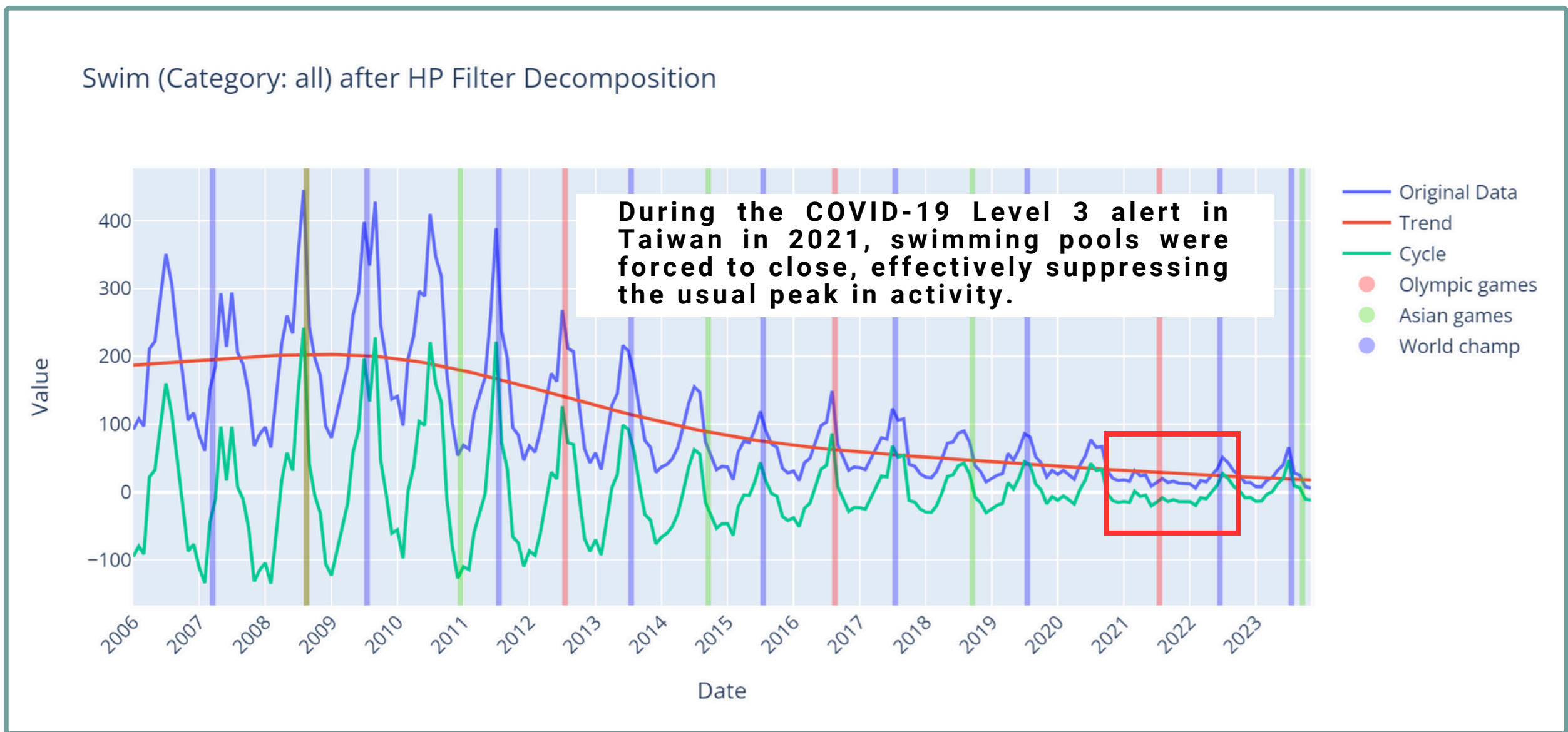
	coef	std err	t	P> t	[0.025	0.975]
Intercept	-4.8008	3.172	-1.514	0.132	-11.058	1.456
Olympics_badminton_dummy	87.7117	17.213	5.096	0.000	53.762	121.674
AsiaGame_badminton_dummy	-16.9457	21.080	-0.804	0.422	-58.531	24.639
WC_badminton_dummy	38.2019	12.449	3.069	0.002	13.644	62.760

Regression of Model 2 (Monthly Basis, Medal)

	coef	std err	t	P> t	[0.025	0.975]
Intercept	-1.6944	3.215	-0.527	0.599	-8.036	4.648
Olympics_badminton_prize	52.6583	15.477	3.402	0.001	22.127	83.190
AsiaGame_badminton_prize	-7.7131	19.711	-0.391	0.696	46.598	31.172
WC_badminton_prize	19.4475	13.385	1.453	0.148	-6.957	45.852

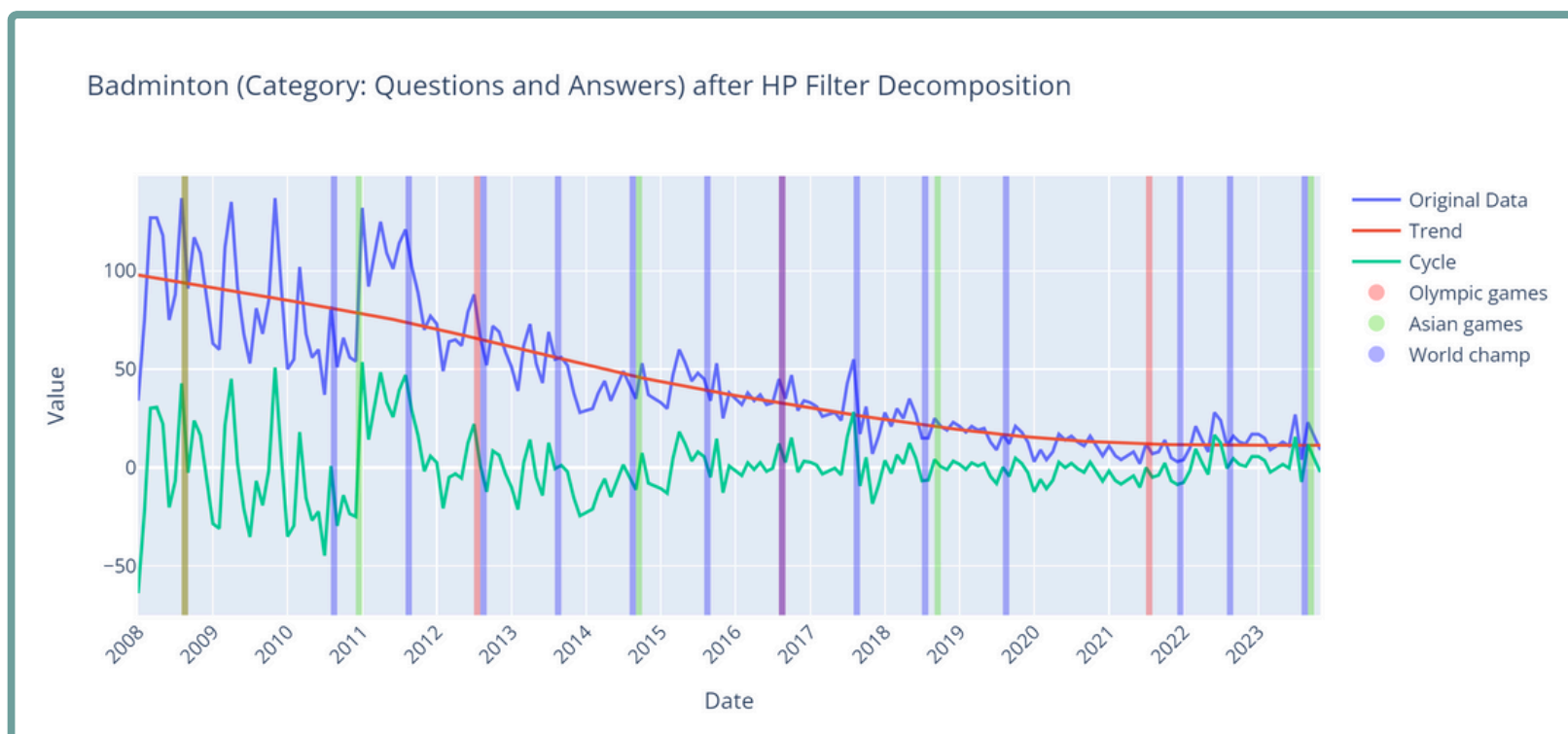
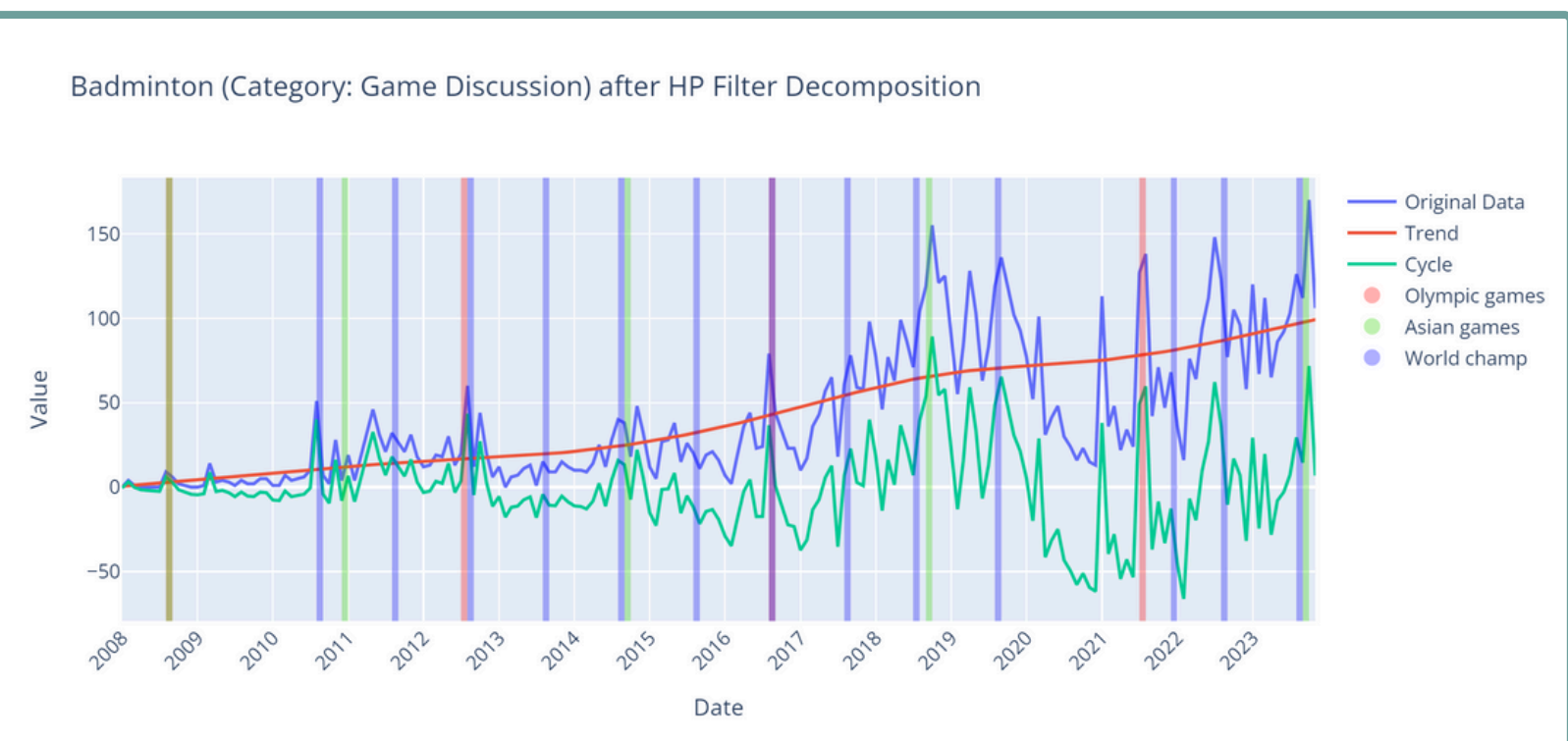
Results for the Olympics and World Championships similarly indicate a significant effect for badminton at the 95% confidence level, particularly in terms of medal counts. Public attention is primarily focused on high-tier competitions, and the performance of our national athletes also generated significant interest.

Swimming – The Seasonal Factor



While major events initially appeared significant, swimming discussions exhibit a strong seasonal cycle that peaks in summer. After controlling for this seasonality, the event effects disappeared. This indicates that public attention is primarily driven by the season and the public's own physical participation in swimming activities, rather than by the tournaments themselves.

Traffic Classification Analysis



To further investigate whether the traffic generated by sports events extends to interest in the sport itself—rather than remaining limited to tournament discussions—we classified the posts by content.

Using badminton as an example, the chart indicates that while event-related posts spiked in alignment with the tournament dates, 'Newbie Q&A' posts (such as beginners asking questions) did not fluctuate with the events, nor did they show any increase after the events concluded.

Conclusion

High-Tier Engagement

Top-tier events drive the most traffic, motivated by both national pride (medals) and genuine interest in the sport itself.

Event Focus

Major single-sport tournaments (e.g., World Championships) often generate more discussion than regional multi-sport games (e.g., Asian Games) due to a more concentrated audience.

Participation Factor

For certain sports (e.g., swimming), public attention is primarily driven by the volume of active participants and seasonal habits, rather than by professional tournaments.